Honest on Average, Not in Every Situation: Situation-Aware Prediction Intervals for 60-Minute CGM Glucose Forecasts

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Abstract

Forecasts of glucose 60 minutes ahead from continuous glucose monitors (CGM) can warn people with type 1 diabetes before glucose becomes dangerously low or high. A forecast is more useful when it comes with a prediction interval, or band, that states how far it can be trusted. Bands are usually judged by their average coverage, the share of true values that fall inside them, and an average can hide failures in particular situations and for particular patients. This initial draft asks whether a nominal 80% band stays honest in each situation, defined as the current glucose level combined with its 30-minute trend. Per-patient forecasters were trained for 215 people from 11 public studies, and eight of the studies (153 patients) were held out for external validation. Pooled over all studies, a standard GARCH band covered between 62% and 91% of outcomes across the situations with enough data. A situation-aware band, which learns a separate width for each situation, kept coverage between 72% and 82% and reduced per-situation miscalibration in all eight held-out studies while producing narrower bands. These results rest on one training seed per patient and are preliminary. Analyses by patient and by day are still in progress.

Status: initial draft for the Midterm Deliverable. The data and core methods are written; the remaining methods and results are planned and marked in red.

Introduction

Type 1 diabetes requires people to manage their blood glucose themselves, every day, with insulin. Too little insulin leads to hyperglycemia, conventionally glucose above 180 mg/dL, which over years damages the eyes, kidneys, nerves and blood vessels. Too much insulin leads to hypoglycemia, glucose below 70 mg/dL, which can cause confusion, seizures and loss of consciousness and can become dangerous quickly (Battelino et al., 2019). Continuous glucose monitors (CGM) measure glucose every five minutes, and that stream of readings makes it possible to forecast where glucose is heading. A forecast 60 minutes ahead is long enough to act on: there is time to eat carbohydrates before a low, or to correct a high with insulin.

A forecast is never exact, so its usefulness depends on how much it can be trusted. A prediction interval, called a band in this paper, gives a range that should contain the true future value with a stated probability. An 80% band is honest, or calibrated, if the true value falls inside it 80% of the time. A band that is too narrow gives false confidence: a person may skip a snack because the band says glucose will stay in range, and then go low. A band that is too wide triggers needless alarms and is soon ignored. Recent glucose forecasters increasingly report bands, using quantile outputs, mixture densities, Monte Carlo dropout, evidential layers or conformal methods (e.g., Sergazinov et al., 2023).

These bands are almost always evaluated by marginal coverage, the share of true values inside the band across all forecasts pooled together. Clinical use, however, is conditional. The cost of a miss depends on the situation: a forecast made at 110 mg/dL while glucose is falling fast matters far more than one made at 200 mg/dL while glucose is steady. A band can reach 80% coverage on average while covering much less in the situations that matter most, because over-coverage elsewhere hides the shortfall. In the same way, a band can be honest for the average patient but not for a particular person, or

honest across a month but not on a given day. In the literature searches for this project, only one glucose prediction study was found that reported band coverage by glucose range, in an intensive care setting, without considering the glucose trend and without proposing a correction [CITE: ICU glucose prediction study, PMC8324237]. No published study applying conformal prediction to CGM forecasting was found, and no study of ambulatory CGM forecasting was found that checks band honesty jointly by situation, by patient and by day. This is recorded as not found in these searches, not as proof that no such study exists.

This paper asks whether a band around a 60-minute CGM forecast stays honest where it is used: in each situation, for each patient, and on each day. A situation is defined as the current glucose level combined with the direction and speed of its change over the previous 30 minutes, for example "between 70 and 120 mg/dL and rising fast." Five levels and five trends give 25 situations. This initial draft addresses the first of the three questions, consistency across situations.

The draft makes three contributions. First, it describes a large evaluation setting: per-patient forecasters for 215 people with type 1 diabetes from 11 public studies, in which three datasets were used to design the methods and the other eight studies, with 153 patients, were held out and used only to check them. Second, it compares a standard band from financial econometrics, GARCH (Bollerslev, 1986), with a situation-aware band that learns a separate width for each situation, a form of group-conditional (Mondrian) conformal prediction (Vovk et al., 2005). Third, it reports initial evidence that the situation-aware band is more consistent across situations than the standard band, including on the held-out studies. Each comparison was judged against a decision rule written down before the analysis was run (Nosek et al., 2018). Because the main analysis uses one training seed per patient, these results are preliminary.

The paper is organized as follows. Section reviews related work on glucose forecasting, uncertainty for glucose forecasts and conformal prediction. Section describes

the 11 studies and how patients and records were selected. Section describes the forecaster, the situations, the bands and the measures of consistency. Section presents the initial results on consistency across situations, and Section briefly discusses them. Analyses by patient and by day will follow in later versions.

Related Work

Glucose forecasting. Short-term glucose forecasting from CGM, insulin and carbohydrate records is an active field, and most work aims at better point accuracy. Recent frameworks such as GLIMMER change how forecasters are trained so that errors in dangerous glucose ranges count more (Khamesian et al., 2026). Public datasets, including OhioT1DM (Marling & Bunescu, 2020), AZT1D (Khamesian et al., 2025) and the MetaboNet consolidation of many public studies (Wolff et al., 2026), now make it possible to test methods across many patients and settings.

Uncertainty for glucose forecasts. A smaller body of work adds bands to glucose forecasts. Gluformer produces a predictive distribution from a transformer (Sergazinov et al., 2023). Other studies use quantile outputs, Monte Carlo dropout or evidential output layers [CITE: Tan and McBeth 2026, arXiv:2603.04955; TFT quantile study, PMC12349322]. One recent study uses uncertainty to shorten the forecast horizon when glucose varies rapidly, which adapts the horizon rather than the band [CITE: MAKE 8:145, 2026, doi:10.3390/make8060145]. Personalization of uncertainty has been approached through meta-learning, which fixes a patient's uncertainty model before deployment rather than updating it during use [CITE: Zhu et al.]. Calibration in this literature is mostly reported as marginal coverage, and a 2026 review of forecasting models for type 1 diabetes calls for calibrated bands and coverage metrics in future benchmarks [CITE: Technologies 14:399, 2026, doi:10.3390/technologies14070399].

Conformal prediction. Conformal prediction builds bands from the errors a model made on held-out calibration data and guarantees average coverage under mild assumptions (Vovk et al., 2005). Conformalized quantile regression combines it with

learned quantiles to produce bands whose width varies with the input (Romano et al., 2019). Deep ensembles offer a different route, using the disagreement between several networks as a measure of uncertainty (Lakshminarayanan et al., 2017).

Group-conditional and adaptive conformal prediction. Mondrian conformal prediction calibrates separately within each predefined group, so that coverage holds within every group and not only on average (Vovk, 2012; Vovk et al., 2005). Later work extends such guarantees to broad classes of overlapping subgroups (Gibbs et al., 2023). For time series, where errors are correlated over time, adaptive conformal inference adjusts the band online after each observed hit or miss (Gibbs & Candès, 2021), and related methods target dynamic series directly (Xu & Xie, 2021). In econometrics, interval forecasts are tested for correct coverage and for clustering of misses (Christoffersen, 1998). These methods exist in general form. What this paper adds is their use with clinically meaningful glucose situations, and evidence across 11 studies on whether the resulting bands stay honest.

[PLANNED: related work on online per-patient band adjustment and on separating real miscalibration from sampling noise in small groups.]

Data

This study uses CGM records from 11 public studies of people with type 1 diabetes, 215 patients in total (Table 1). A CGM is a small sensor worn on the body that reports interstitial glucose every few minutes. The studies have two roles. Three datasets, with 62 patients, were used to develop and tune the methods. Eight further studies, with 153 patients, were never used to design or tune any method. They were used only afterwards, to test methods already fixed on the development data, as an external validation. This separation matters because a band rule tuned and judged on the same patients can look better than it is.

Development Datasets

AZT1D (Khamesian et al., 2025) contains real-world records from 25 adults using automated insulin delivery (AID), also called closed-loop therapy, in which a pump adjusts

insulin automatically from sensor readings, with a Dexcom G6 sensor. It was downloaded from Mendeley Data under a CC BY 4.0 license.

OhioT1DM (Marling & Bunescu, 2020) contains 12 adults on sensor-augmented pump (SAP) therapy, in which the pump delivers insulin as programmed and the sensor informs the user but does not steer the pump, with a Medtronic Enlite sensor. The dataset is normally released under a data use agreement. A direct access request received no reply, so the 12 patients were taken from the MetaboNet consolidation described below.

HUPA-UCM (Alvarado, 2024) contains 25 adults on pump or multiple daily injection (MDI) therapy with a FreeStyle Libre 2 sensor, with short records of about 13 days each. It was downloaded from Mendeley Data under a CC BY 4.0 license, and the preprocessed version distributed with the dataset was used.

Held-Out External Studies

The external studies come from the public subset of MetaboNet (Wolff et al., 2026), which brings 14 public type 1 diabetes datasets into one harmonized file on a common 5-minute grid. Its public subset is released for non-commercial research. Studies hosted by the Jaeb Center for Health Research follow Jaeb terms of use, and the others are under CC BY 4.0 (Jeffries et al., 2026). Eight studies were chosen to span therapy, sensor and age: Loop, DCLP3, DCLP5, PEDAP, Flair, IOBP2, ReplaceBG and BrisT1D. The MetaboNet public subset holds only the closed-loop arms of the randomized trials DCLP3, IOBP2 and Flair.

Within each study, patients with at least 45 days of CGM readings were eligible. Eligible patients were taken in a random order, and for each one the 45-day stretch with the most readings was kept. A patient was accepted if that stretch had at least 90% of the expected readings, until 20 patients per study were reached. Only 13 BrisT1D patients met these rules, which gives 153 patients overall. Their ages range from 2 to 74 years, and 69 of the 153 are under 18. The patient counts, ages, therapy labels and sensors in Table 1 come from the metadata file written when this panel was built; the age ranges describe the

sampled patients, not the whole studies.

Table 1

Studies Used in This Paper

Study	Patients	Ages (years)	Therapy	Sensor	Role
AZT1D	25	Adults	AID	Dexcom G6	Development
OhioT1DM	12	Adults	SAP	Medtronic Enlite	Development
HUPA-UCM	25	Adults	SAP or MDI	FreeStyle Libre 2	Development
Loop	20	2–65	AID	Dexcom G6/G5 (1 Libre)	Held out
DCLP3	20	14–71	AID	Dexcom G6	Held out
DCLP5	20	8–13	AID (3 SAP)	Dexcom G6	Held out
PEDAP	20	2–5	AID	Dexcom G6	Held out
Flair	20	15–27	AID	Medtronic Guardian 3	Held out
IOBP2	20	9–74	AID	Dexcom G6	Held out
ReplaceBG	20	26–74	SAP (1 unrecorded)	Dexcom G4 (1 unrecorded)	Held out
BrisT1D	13	18–24	12 of 13 unrecorded	Dexcom G6/One, Guardian 4	Held out
Total	215				

Note. AID = automated insulin delivery (closed loop); SAP = sensor-augmented pump; MDI = multiple daily injections. Ages for held-out studies are the range among sampled patients.

Signals and Preprocessing

All records are on a 5-minute grid: AZT1D and the preprocessed HUPA-UCM files are distributed this way, and MetaboNet places every study it holds on a common 5-minute grid. Four recorded signals were used: CGM glucose, basal insulin (the background rate the pump delivers, in units per hour), bolus insulin (doses given for meals or corrections) and carbohydrate intake. Other signals, such as heart rate and steps in HUPA-UCM, were not used.

Preprocessing followed the same rules for every study. Basal insulin is logged only when the rate changes, so within each patient it was carried forward to each 5-minute step. Missing bolus and carbohydrate entries were set to zero, meaning that nothing was given or eaten. Rows without a glucose reading were dropped rather than filled with an invented

value. For the held-out studies, gaps in glucose of 30 minutes or less inside the 45-day stretch were filled by linear interpolation, and rows in longer gaps were dropped. Some held-out patients had no basal record at all (10 of the 20 PEDAP patients, all 13 BrisT1D patients and one ReplaceBG patient). Their basal input was set to zero, and this is flagged in the panel metadata.

Each patient's record was cut into forecast windows of two hours of history and a target 60 minutes ahead. The windows were split in time order for each patient: the first 64% for training, the next 16% for validation and the last 20% for testing. The validation part is used for early stopping and to calibrate every band; the test part is used only for scoring. Because the split is chronological, no later data reaches the model or the band during fitting or calibration.

Caveats

Three properties of the data shape how results are reported. First, pooled numbers can be driven by one patient: in HUPA-UCM, one patient accounts for 53% of the pooled rows. For this reason results are reported per patient and per study, not only pooled over readings. Second, about 8% of pooled HUPA-UCM readings are straight-line filler, runs that look like interpolation rather than measured values. They were not removed in the analyses reported here [CHECK: confirm the filler was never excluded]. Third, therapy labels in the held-out studies are taken as recorded by MetaboNet, and they are missing for 12 of the 13 BrisT1D patients and one ReplaceBG patient.

[PLANNED: an expanded held-out panel of up to 60 patients per study, built with the same selection rules, is being analyzed; all counts above refer to the 20-per-study panel.]

Methods

This section describes the forecaster that produces the point predictions, the grouping of forecasts into situations, the bands compared, and the way their consistency across situations is measured. Throughout, a band is *calibrated* in a group of forecasts when its *coverage*, the share of true values that fall inside it, matches the stated

probability, here 80%.

Forecaster

Each patient received a separate forecasting model, following the per-patient design of GLIMMER (Khamesian et al., 2026). The model is a convolutional-recurrent network (CNN-LSTM): three one-dimensional convolutional layers (16, 32 and 16 filters, kernel size 3, ReLU activations) extract short local patterns from the input window, and a single long short-term memory (LSTM) layer with 8 units summarizes the window into one forecast. The network is small, with 4,265 trainable parameters.

The input is the 2-hour window of 24 readings described in Section . Each time step carries six variables: the four recorded signals (CGM, basal insulin, bolus insulin and carbohydrate intake), a moving average of the last 200 CGM readings (about 17 hours), and a categorical glycemic region (below 70, 70 to 180, or above 180 mg/dL). Inputs are standardized with statistics from the training windows. The target is the CGM reading 60 minutes (12 steps) after the last input reading.

Models were trained on the training windows with the Adam optimizer (learning rate 10^{-3}, batch size 64) on mean squared error for up to 30 epochs, keeping the epoch with the lowest validation root mean squared error and stopping after 5 epochs without improvement. Each patient's model had its own random seed, so that no two patients' models share the same random initialization. The main analysis uses one seed per patient.

[PLANNED: robustness forecasters (a CNN-Transformer, a 30-minute horizon, a second seed for every development patient, and one pooled forecaster trained only on the external studies), one sentence each.]

Situations

The central idea of this paper is that forecast uncertainty depends on the *situation*: where glucose currently is and where it is heading. A situation is the combination of the current glucose level and the 30-minute trend, both known at the moment the forecast is issued.

The level g_t is the current CGM reading, placed into one of five ranges: below 70 mg/dL (hypoglycemia), 70 to 120, 120 to 180, 180 to 250, and 250 mg/dL and above, where each range includes its lower edge. The trend is the change over the previous six readings (30 minutes at 5-minute sampling), $\Delta_t = g_t - g_{t-6}$, placed into one of five categories with edges at -30, -10, 10 and 30 mg/dL: falling fast ($\Delta_t < -30$), falling slowly ($-30 \le \Delta_t < -10$), flat ($-10 \le \Delta_t < 10$), rising slowly ($10 \le \Delta_t < 30$), and rising fast ($\Delta_t \ge 30$). Crossing the five levels with the five trends gives 25 situations,

$$s_t = 5\,\ell(g_t) + k(\Delta_t), \qquad \ell, k \in \{0, \ldots, 4\}, \tag{1}$$

where ℓ indexes the level range and k the trend category. Windows issued before six readings of history were available were dropped. Some situations are common (flat glucose between 120 and 180 mg/dL) and some are rare (glucose below 70 mg/dL and rising fast), which matters for how reliably each can be calibrated.

Bands

All bands are calibrated on the validation windows and scored on the test windows at a nominal coverage of 80%, so that the miscoverage rate is $\alpha = 0.2$. Every band is clipped to the 40 to 400 mg/dL reporting range of the sensor. Let $\hat{y}_t$ denote the point forecast and $e_t = y_t - \hat{y}_t$ the forecast error (residual).

Baseline: GARCH band. The generalized autoregressive conditional heteroskedasticity model, GARCH(1,1) (Bollerslev, 1986), comes from finance, where it tracks how the size of errors changes over time. Its variance follows

$$\sigma_t^2 = \omega + a\,(e_{t-1} - \mu)^2 + b\,\sigma_{t-1}^2, \tag{2}$$

with parameters μ, ω, a, b fitted on each patient's validation residuals. Because a forecast error is only known once its target time has passed, the variance is filtered with a lag equal to the 60-minute horizon and projected 12 steps ahead. The band is the Gaussian interval

$$\hat{y}_t + \mu \pm z_{0.9}\,\sigma_t, \qquad z_{0.9} \approx 1.28. \tag{3}$$

The GARCH band is used as fitted, with no further rescaling, so its average coverage depends on how well the Gaussian assumption holds for each patient.

Split conformal band. Conformal prediction (Vovk et al., 2005) sizes the band from past errors without assuming a distribution. With n validation scores $r_i = |e_i|$ sorted in increasing order, the half-width is

$$\hat{q} = r_{(\lceil (n+1)(1-\alpha) \rceil)}, \tag{4}$$

the $\lceil (n+1)(1-\alpha) \rceil$-th smallest absolute validation error, and the band is $\hat{y}_t \pm \hat{q}$. When validation and test windows are exchangeable, this guarantees at least $1-\alpha$ coverage on average over all forecasts, but not within any particular situation. The quantile was computed either from all patients pooled or from each patient alone.

Situation-aware band. The situation-aware band applies the same conformal rule separately within each situation, a form of grouped calibration known as Mondrian conformal prediction (Vovk et al., 2005). For each situation s, the half-width $\hat{q}_s$ is the conformal quantile of the absolute validation errors of the windows in that situation, and a test forecast issued in situation s_t receives the band

$$\hat{y}_t \pm \hat{q}_{s_t}. \tag{5}$$

Validation errors are pooled across the patients of a dataset within each situation, so that rare situations still have enough data. A situation with fewer than 100 validation windows falls back to the pooled quantile $\hat{q}$. Under exchangeability within each situation, this construction gives the coverage guarantee separately in every situation that has its own quantile, rather than only on average. Whether that guarantee holds up on real CGM data, where windows are not exchangeable, is what the results test.

[PLANNED: (1) a GARCH-normalized situation band, in which absolute errors are divided by σ_t before the per-situation quantile; (2) an online per-patient scale, in which the situation half-width is multiplied by $\exp(\theta_t)$ and each patient's θ_t starts at 0 and is updated once the outcome is known, one horizon later, by $\theta_{t+1} = \theta_t + \eta\,(\text{miss}_{t-12} - \alpha)$ with

$\eta = 0.01$; (3) a cold start for new patients, with situation quantiles computed from other patients only; (4) comparison baselines: deep ensembles (Lakshminarayanan et al., 2017), conformalized quantile regression (Romano et al., 2019) with and without a per-situation correction, and adaptive conformal inference (Gibbs & Candès, 2021).]

Measuring Consistency

A band can be calibrated on average while covering too little in some situations and too much in others. This paper calls a band *consistent* when its coverage is close to the nominal 80% in every situation, and measures consistency as follows.

Coverage per situation. For situation s with n_s test windows, coverage is

$$\hat{c}_s = \frac{1}{n_s} \sum_{t:\, s_t = s} \mathbf{1}\{y_t \in [L_t, U_t]\}, \tag{6}$$

where L_t and U_t are the lower and upper ends of the band.

Calibration error across situations. The summary of consistency is the average absolute distance from the nominal 80% across situations,

$$\mathrm{CE} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} |\hat{c}_s - 0.8|\,, \tag{7}$$

reported in percentage points, where $\mathcal{S}$ contains the situations with at least 30 test windows. A value of 0 means every situation is covered exactly as claimed. A companion measure, the *dispersion*, replaces 0.8 with the band's own overall coverage $\hat{c}$; it ignores a uniform shift in coverage between the validation and test periods and measures only how unevenly the band covers different situations.

Noise floor. Even a perfectly consistent band would show some spread in $\hat{c}_s$, because each situation contains a finite number of windows. This spread is larger than an independent-sample formula suggests, because consecutive forecast windows overlap and misses cluster in time. To estimate the dispersion that sampling noise alone would produce, the situation labels of each patient's test windows are circularly shifted by a random lag of at least one day (288 windows), and the dispersion is recomputed against

the unshifted sequence of hits and misses; patients with fewer than two days of test windows are left unshifted. A circular shift keeps the time structure of both the labels and the misses but breaks the link between them, so any remaining spread is chance. The *noise floor* is the mean dispersion over repeated shifts (30 to 50 in the reported analyses), and the *excess* is the observed dispersion minus the floor. An excess near zero means the remaining unevenness of a band cannot be distinguished from sampling noise.

Uncertainty and decision rules. The patient, not the forecast window, is the unit of analysis, since one patient contributes thousands of closely related windows. Uncertainty in a metric is therefore estimated with a patient-level cluster bootstrap: patients are resampled with replacement 1,000 times, the metric is recomputed on all windows of the sampled patients, and the 5th and 95th percentiles give a 90% interval. Each comparison between bands was registered with its metric, threshold and data before it was run.

[PLANNED: further analyses, one sentence each: consistency by patient and by day; error-size predictability (how much of the size of each error can be predicted from situation, patient and richer context with gradient boosting); patient difficulty (regression of log patient error on glucose variability and on the variability of 30-minute changes); and the share of error attributable to the sensor.]

Initial Results: Consistency of the Bands Across Situations

This section reports one initial result: whether an 80% band is honest in every situation, not only on average, using the situations of Section and the calibration error of Section . Each forecaster was trained with a single random seed, so all numbers below are preliminary.

Coverage by Situation

The standard GARCH band was honest on average but not by situation. On AZT1D, its coverage ranged from 64% (70 to 120 mg/dL, rising fast) to 95% (below 70 mg/dL, falling fast). Pooled over all 11 studies, the range was 62% to 91% (Figure 1, left).

The band was too narrow when glucose rose quickly from the normal range and too wide when glucose was already low and falling, both situations in which a warning could matter.

The situation-aware band was more even. Pooled over 11 studies, its coverage stayed between 72% and 82% in every situation with enough data, and between 78% and 82% in 23 of those 24 situations (Figure 1, right). The one exception, 72%, was glucose below 70 mg/dL and rising slowly.

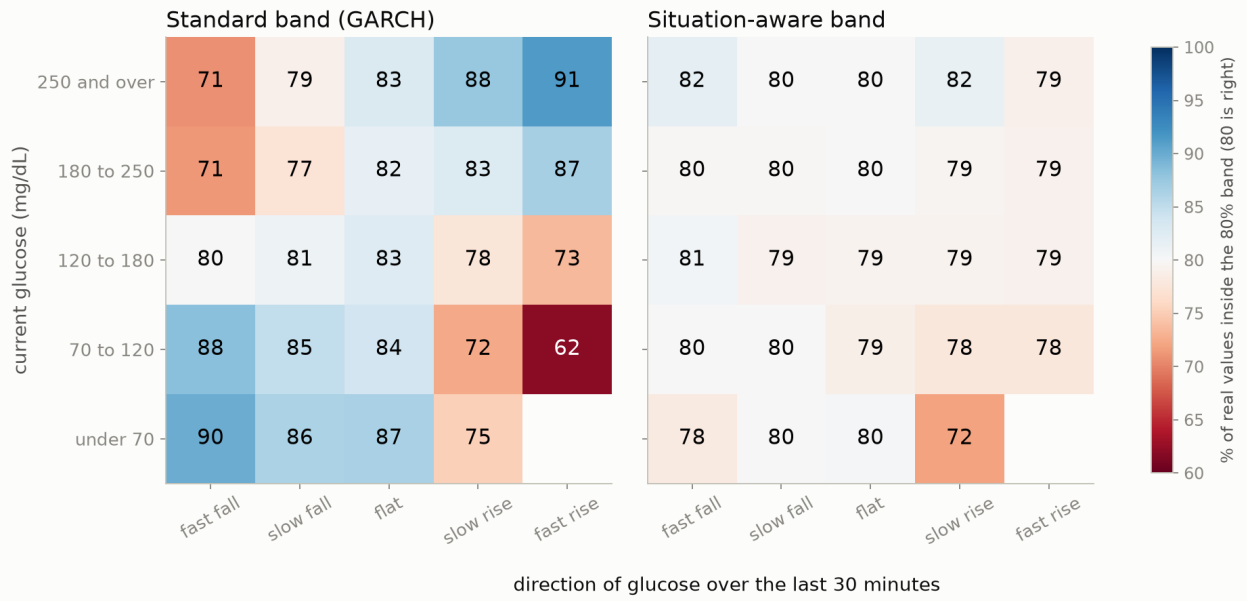


Figure 1

Share of true glucose values inside the 80% band in each situation, pooled over 11 studies. Left: the standard GARCH band (62% to 91%). Right: the situation-aware band (72% to 82%). Rows are the current glucose level and columns the 30-minute trend; a value of 80 is on target. The empty cell (below 70 mg/dL, rising fast) had fewer than 100 test windows and is not shown.

External Validation

Because the method was designed on the three development datasets, it was then tested on the eight held-out MetaboNet studies (153 patients). The rule for success was written down before the run: the situation-aware band had to show at least 20% lower

calibration error across situations than GARCH in at least six of the eight studies. The rule was met in all eight studies (Figure 2). The situation-aware band was 2.3 to 4.2 percentage points from 80% on average, against 5.5 to 6.8 points for GARCH, a reduction of between 29% and 67% depending on the study. Overall coverage stayed close to the target (78.5% to 81.2%), and the bands were narrower than GARCH in every study, by about 8 to 12 mg/dL in average full width (84 to 99 mg/dL against 94 to 108 mg/dL).

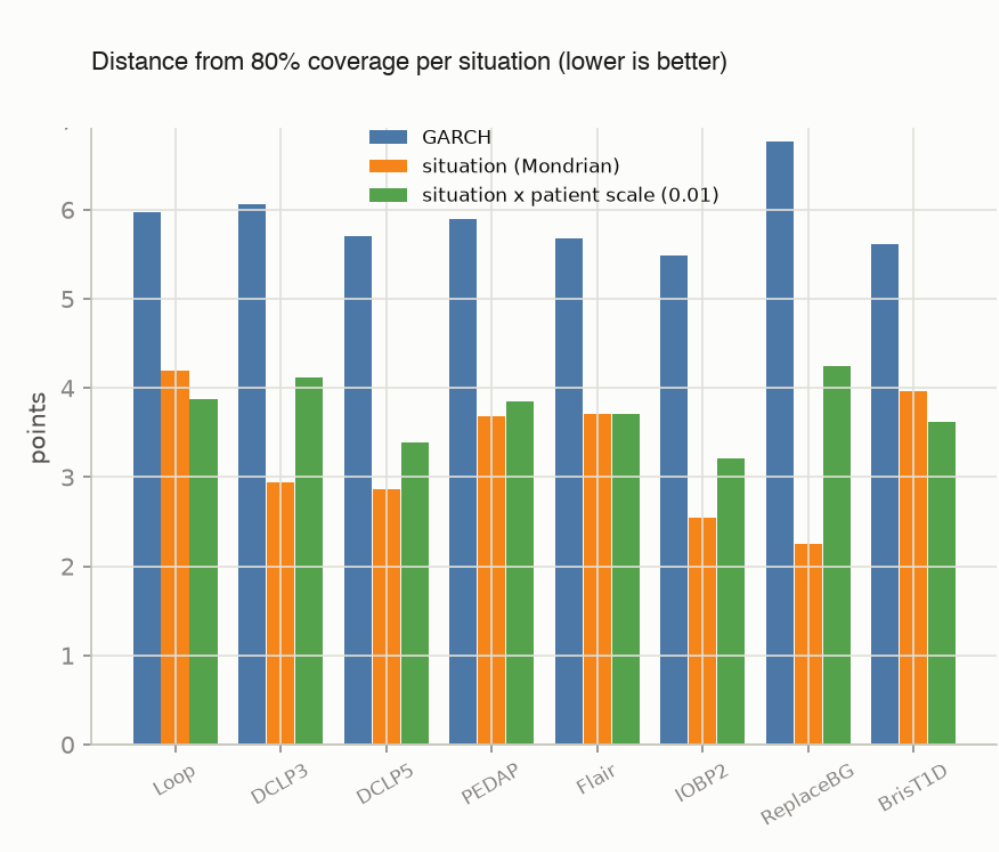


Figure 2

External validation on eight held-out MetaboNet studies (13 to 20 patients each): average distance from 80% coverage across situations, in percentage points, for GARCH (blue), the situation-aware band (orange) and the situation-aware band with a per-patient adjustment (green, a planned result). Lower is better.

Planned Results

[PLANNED: (1) the noise floor, which separates real miscalibration from sampling noise in autocorrelated forecast windows; (2) consistency across patients and the online per-patient adjustment; (3) consistency across days; (4) how predictable the size of the error is; (5) why some patients are harder to forecast than others; (6) the share of error that is unavoidable given sensor noise; (7) the trade-off between consistency and catching large, dangerous errors; (8) a second-seed check and the effect of the situation grid size.]

Discussion

The initial result suggests that a band can be honest on average while being systematically too narrow or too wide in specific situations, and that sizing the band by situation removes most of this unevenness without widening it. Because the gain held on eight studies never used to design the method, it is less likely to reflect tuning to the development data. The main limitation is that a band that is honest in every situation does not necessarily catch more dangerous forecast errors: consistency is a property of honesty, not of clinical benefit. The results also come from a single training seed per patient and remain preliminary.

[PLANNED: full discussion, including clinical relevance, comparison with prior work, and limitations of the data (45-day external records, 13 to 20 patients per study).]

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Process Documentation: Midterm Deliverable

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CP193: Capstone Directed Study I (Machine Learning)

Prof. Watson

October 31, 2026

Abstract

This document describes how I carried out the first two months of my capstone on glucose forecasting for type 1 diabetes, from early September to late October. The project began as a replication of GLIMMER, a published study that trains forecasters to weight errors in dangerous glucose ranges more heavily. A bug in that replication erased its main gain, and my first uncertainty bands raised far too many false alarms. Those turns led to the question I study now: whether the band around a 60-minute forecast stays honest in each glucose situation, for each patient and on each day. I explain how I came to define success, how a written research protocol and a register that keeps every failed hypothesis guide my testing, and how I checked results on eight outside studies never used to design the methods. I also describe my planning against what actually happened, my use of an AI coding assistant and the checks I place on it, how I used my advisor's feedback, and the work that remains before the Full Draft.

Overview

My capstone is now a study of how honest the uncertainty around a glucose forecast is. People with type 1 diabetes try to keep their glucose between two lines. Below 70 mg/dL is hypoglycemia, glucose that is too low, which can turn dangerous quickly. Above 180 mg/dL is hyperglycemia, glucose that is too high, which harms the body when it lasts. A forecast that looks 60 minutes ahead matters because an hour is enough time to act, for example by eating or by adjusting insulin, before the reading crosses either line.

A forecaster predicts glucose 60 minutes ahead, and a band around the forecast tells the person how far to trust it. An 80% band is honest if the true value falls inside it 80% of the time. The question I am answering is whether the band stays honest where it matters: in each situation (the current glucose level plus the direction it has moved over the last 30 minutes), for each patient, and on each day. I tested this on 215 patients from 11 public studies. Three studies were used to design the methods. The other eight (153 patients) were held out and used only to check them.

The project started as a replication of GLIMMER (Khamesian et al., 2026), a published study that weights a forecaster's mistakes in dangerous glucose ranges more heavily. The following sections explain how I got from there to here: the turns and dead ends, how I defined success, how I used generative AI, how I used feedback, and what remains before the Full Draft. The work product reports the results; this document is about how I reached them.

Two parts of the process are the ones I am proudest of. The first is catching my own errors. Several times a result looked like progress until I checked it again, and each time I treated the correction as part of the work rather than something to hide. The second is the deep understanding of the material I gained along the way. I started with little knowledge of diabetes data or of how uncertainty is measured, and I now understand where forecast errors come from, which ideas from finance, weather and statistics carry over to glucose and which do not, and why a band that looks fine on average can fail the people who need

it most.

Navigation: How the Work Moved

This section follows the project from my choice of topic in early September to the research program I run now. It records the turns and the dead ends, because each one changed what I did next.

Starting Point and the GLIMMER Replication

At the start of the semester I was debating between projects that would support my career path and projects I found interesting. I concluded that I would rather work on something I enjoy and can happily pour my time into, so I chose this one.

The first step was finding data. The standard dataset, OhioT1DM (Marling & Bunescu, 2020), requires an access request. I requested access, and the dataset's maintainer granted it about five days later, in mid-September. In the meantime I had started with AZT1D (Khamesian et al., 2025), a newer dataset of 25 patients. I also found OhioT1DM inside MetaboNet (Wolff et al., 2026), a public collection of diabetes datasets, and I checked MetaboNet row by row against AZT1D's own files before relying on it.

I began by replicating GLIMMER (Khamesian et al., 2026), a 2026 study from the group that built AZT1D. It trains forecasters to weight mistakes in dangerous glucose ranges more heavily, and it tunes that weighting for each patient. In my replication, the weighted models caught more dangerous events (the share caught rose from 0.507 to 0.722), but overall accuracy was worse.

Then I found a bug: each patient's weight search had started from the same random seed, so every search was identical. After the fix, the gain from per-patient tuning disappeared (a typical error of 41.12 mg/dL for the tuned model against 41.18 mg/dL for fixed weights). This taught me to dive deeper into whatever I analyze, rather than trusting a result at first sight. Since then I re-check results before relying on them.

After my first presentation, Prof. Watson suggested forecasting the whole glucose curve step by step, which is called recursive forecasting. The model predicted the next 5

minutes, fed its own prediction back in as if it were real data, and repeated this to build the curve over the next several hours. To keep going on its own, the model had to predict every input together: insulin, meals and glucose. The curve came out as a nearly straight line, because the model always guessed that no meal was coming. Adding the time of day as a fixed input, since meals cluster at certain hours, did not help. When I gave the model the real logged meals instead of asking it to guess them, the curve improved and rose with the meal, but this was on one patient and only because the model was told the meals in advance. The approach could not predict meals, which is the part a person would need, so I set it aside. I wrote at the time that my next steps were unclear to me.

From Forecasting to Uncertainty

Building a better prediction model is a crowded space, with many groups competing on accuracy. I decided to expand into something more niche: the uncertainty around a forecast. I thought I had a better chance of finding a gap in the research there. So in late September I moved from improving the forecast to estimating how far to trust it.

The idea was to put a band around each forecast and use it as a warning. If the band's edge crossed into the dangerous range, meaning its lower edge fell below 70 mg/dL or its upper edge rose above 180 mg/dL, it triggered a warning (Figure 1). A wide band warns often, and a narrow band warns rarely.

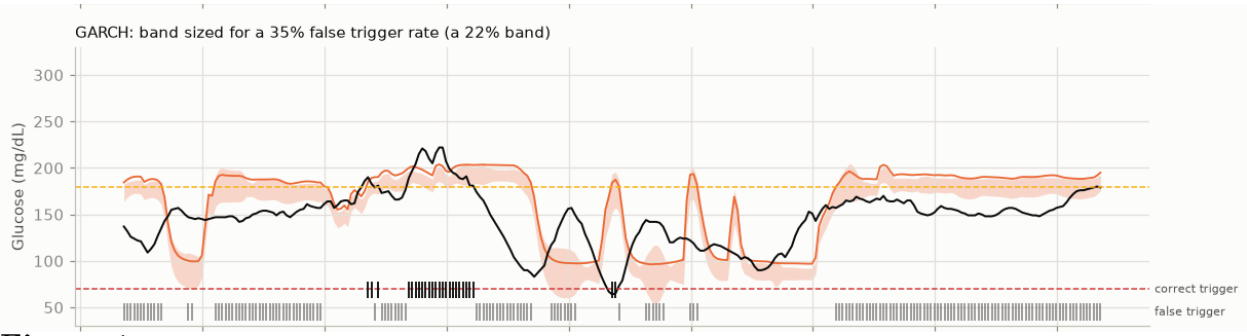


Figure 1

One patient's day: the actual glucose, the GARCH band, and the warnings it raised. Black ticks are correct warnings and grey ticks are false alarms.

To judge the forecasts and their warnings I used the Clarke error grid (Clarke et al., 1987), a standard chart that sorts glucose estimates by how much harm an error could cause (Figure 2). Each forecast is one point, with the actual glucose on one axis and the forecast on the other. Zone A is clinically accurate. Zone B is off but harmless. Zone C would lead to unnecessary treatment. Zone D is a dangerous miss: the person is actually low or high, but the forecast says normal. Zone E points the wrong way, calling a low a high or the reverse. Throughout this document, dangerous readings means real lows and highs, and zone D is what a warning system must avoid.

I borrowed band methods from other fields: GARCH (Bollerslev, 1986), which tracks changing volatility in financial markets; the analog ensemble (Delle Monache et al., 2013), which builds a weather forecast from similar past situations; and conformal prediction (Vovk et al., 2005), a statistical method that sizes the band from past errors.

The bands failed my first test. Ordinary 80% and 95% bands raised alarms on 68% to 100% of safe readings. When I compared the methods at the same false-alarm rate, they caught between 74% and 76.5% of dangerous readings, so none was clearly better (Figure 3). I then added three more methods: EWMA, another volatility tracker from finance; locally weighted conformal prediction, which sizes the band from errors on similar windows; and time-of-day bands, which use past errors at the same hour. This did not change the picture: all six alarm curves were nearly identical.

In late September I looked at what widening the band actually did on the Clarke grid (Figure 4). Making the GARCH band wider pushed more forecasts into zone C, the false alarms, from about 0.1% to 2.5%, while zone D, the dangerous misses, barely dropped. A wider band mostly added alarms. I wrote in my journal: “I don’t think that I would be able to actually create meaningful insights without increasing the false hyper/hypoglycemic rates drastically.”

Instead of testing more methods, I then looked at where the error came from. More data would barely help, and the model did not degrade over time. What did stand out was



Figure 2

The Clarke error grid, with zones A to E labelled. Each dot is one 60-minute forecast for one AZT1D patient, placed by the actual glucose and the forecast.

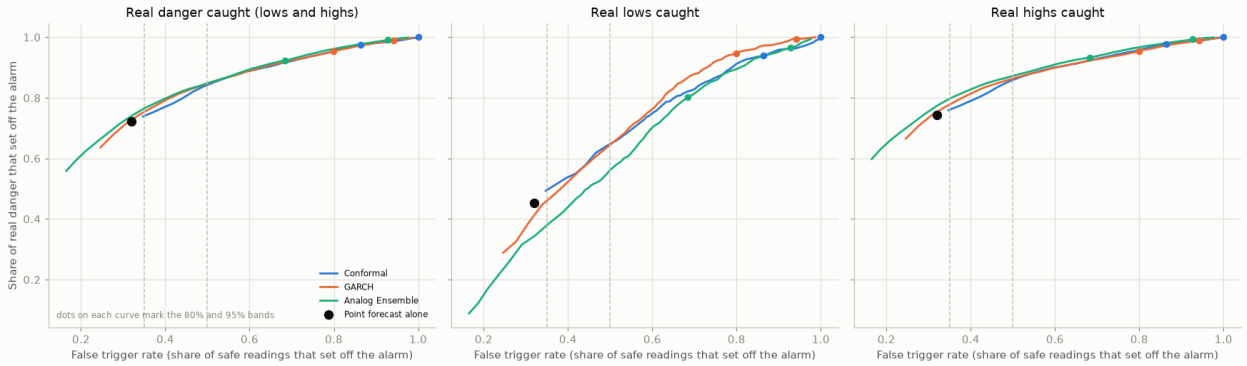


Figure 3

The share of real dangerous readings that set off the alarm against the share of safe readings that set it off, for all danger, lows and highs. Every method lands on nearly the same curve.

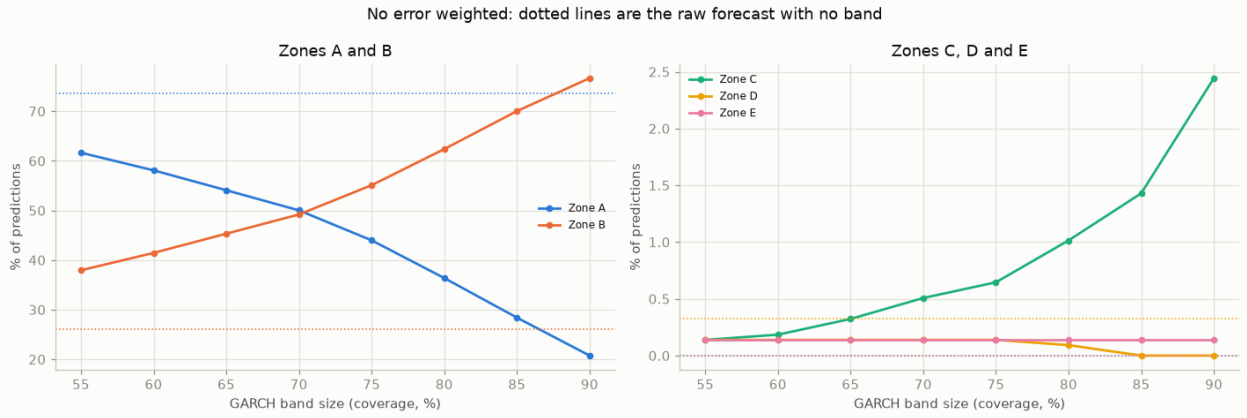


Figure 4

Widening the GARCH band from 55% to 90% coverage. Left: the share of forecasts in Clarke zones A and B. Right: the share in zones C, D and E. Zone C rises from about 0.1% to 2.5% while zone D falls only slightly. Dotted lines are the plain forecast with no band.

that the band was honest on average but not in every situation, which pointed to the direction I took next.

Choosing One Direction

The direction I am pursuing is a situation-aware band. A situation is the current glucose level plus the direction it has moved over the last 30 minutes, for example "between 70 and 180 mg/dL and falling fast." Each situation gets its own band width, learned from past errors in that situation. Statisticians call this kind of grouped calibration Mondrian conformal prediction (Vovk et al., 2005).

It made the bands far more consistent across situations (Figure 5). Pooled over 11 studies, the standard GARCH band's coverage ranged from 62% to 91% across the 25 situations, while the situation-aware band ranged from 72% to 82%, and mostly from 78% to 82%. It also held on data it was never designed on. On OhioT1DM, with the rule for success written down first, the distance from the 80% target fell by 41%, 9 of 12 patients were closer, and the bands were 12% narrower. It did not catch more of the large errors, so the gain is in honesty, not in catching more danger.

A second avenue came from Prof. Watson's written feedback on my early proposal, which suggested approximate entropy, autoencoders, and separating changes driven by meals and insulin from subtler ones. In early October I tested the first two. Sample entropy added little, and the autoencoder mostly found glucose level and trend again, the same structure as the situations I was already using. Neither showed anything new, so I closed this line. The section on feedback describes each test.

From Exploration to a Research Program

By mid-October I had many small results and no clear way to know which were real. I tried a simulation as an answer key, with simulated patients whose true answer I knew. The simulator produced patients that were too easy to forecast (a typical error of 11 to 16 mg/dL, against 31 to 39 mg/dL on real patients), so it could not answer my question, and I decided to drop the simulated patients.

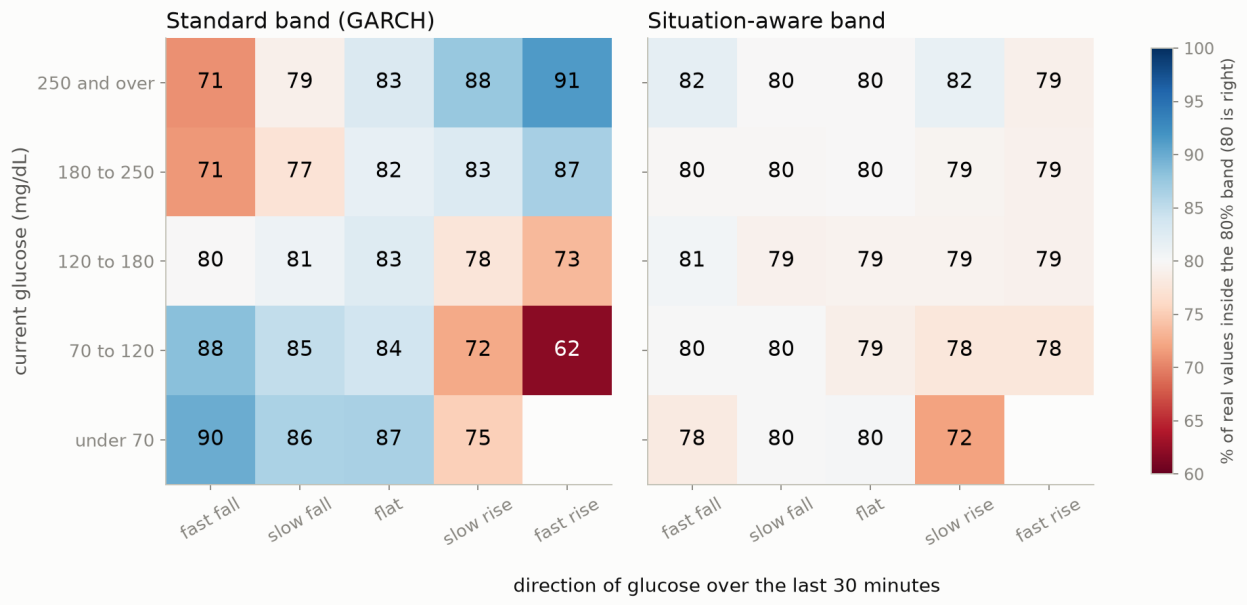


Figure 5

The share of real values inside the 80% band in each situation, for a standard GARCH band and the situation-aware band, pooled over 11 studies. A value of 80 is right.

Instead, I moved from exploration to structured testing on far more data: eight more public studies from MetaboNet, held out from design and used only to check the methods, with a rule for success written before each test. The section on the use of generative AI describes the protocol behind that testing.

Planning: The Timeline I Set and What Happened

At first I kept checklists in Notion. They worked for single tasks, but they were not good enough to see the whole semester at once. In early October I drew a week-by-week plan by hand on my iPad: a square for each remaining week, with its course deadlines and the deliverables I set for myself (Figure 6), and a task list for that day (Figure 7). I then realized that handwriting was slow and could not be searched or read by AI, so I moved the plan into a Gantt chart in Notion. Its fixed points are my second-reader meeting, the Midterm at the end of October, the Second Presentation with a go or no-go decision in late November, and the Full Draft in mid-December. Table 1 compares the plan with what

happened.

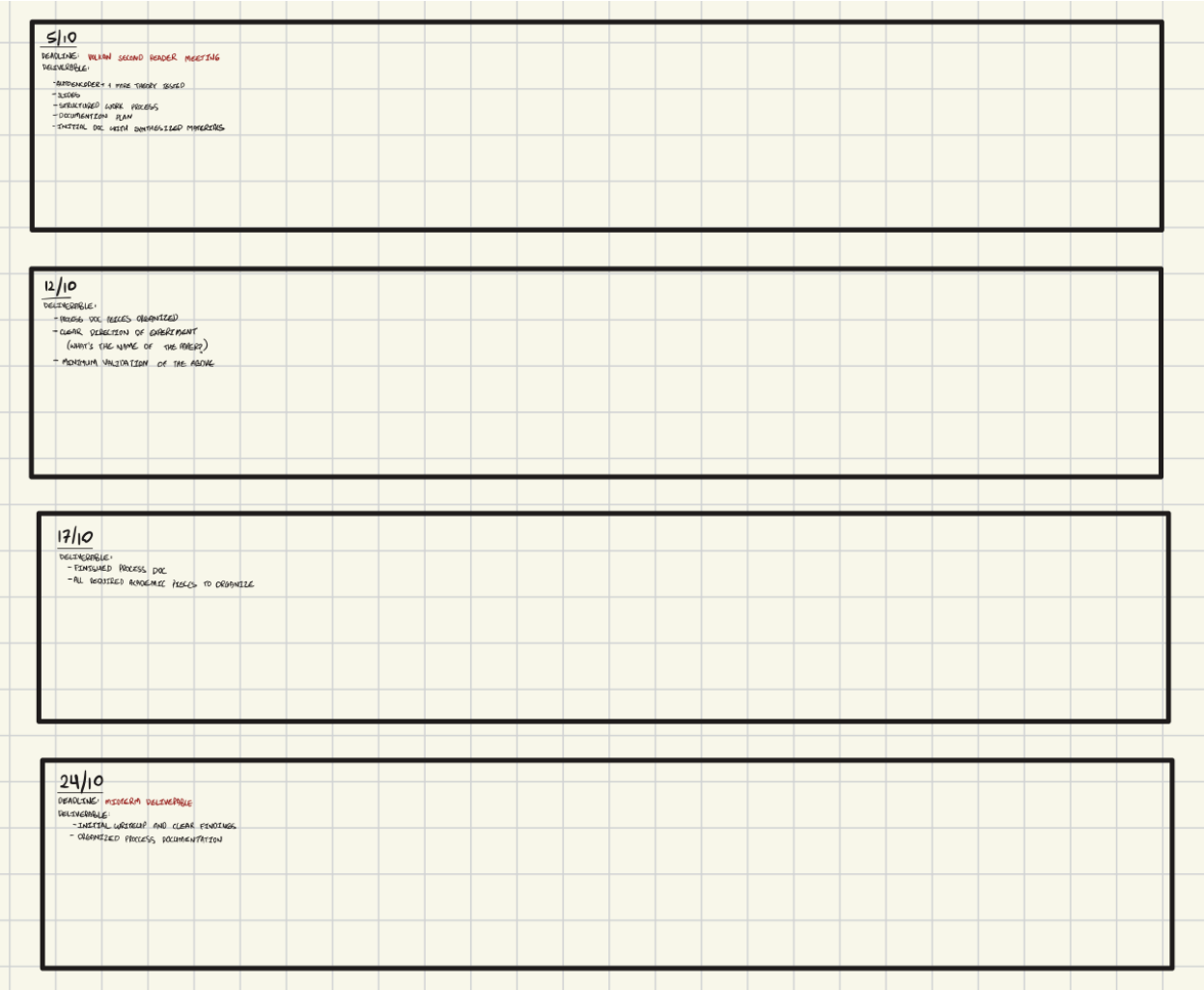


Figure 6

My hand-drawn plan from early October: one square per remaining week, with course deadlines and my own deliverables for the first four weeks.

The pattern is clear. The research ran ahead of the plan, and the documentation fell behind it. [TO FILL: how I plan to rebalance research and writing between now and the Full Draft]

Working Habits and Tools

I block 7:30 to 9:30am for the capstone. In early October I noticed that I skipped this block on some days, and I listed three options: accept three days a week, wake up

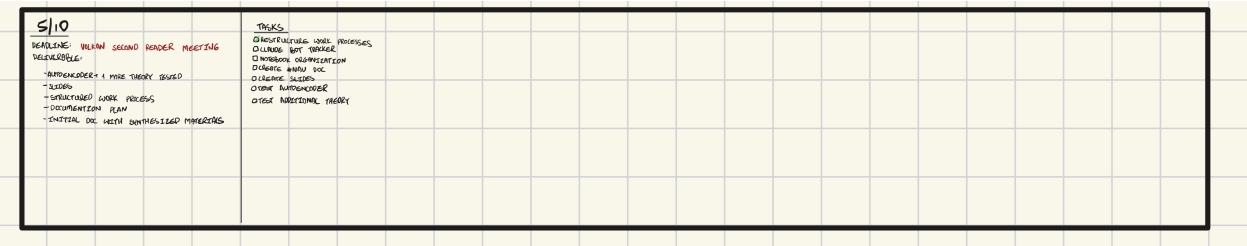


Figure 7

My hand-drawn task list from the same day in early October, when I reorganized the project.

Table 1

Deliverables I Planned in Early October Compared With What Happened

Week of	Planned	What happened
Oct 5	Autoencoder and one more theory tested; slides; structured work process; documentation plan; first document of synthesized materials	Autoencoder and entropy tested, then closed; slides ready for my second-reader meeting
Oct 12	Process documentation pieces organized; a clear direction ("what's the name of the paper?"); minimum validation	Direction and validation went beyond the plan: a written research protocol and tests on 11 studies; process pieces not yet organized
Oct 17	Finished process documentation; academic pieces organized	A paper draft built from the findings; research and corrections continued; a lost data file stopped one run; process documentation slipped
Oct 24	First write-up with clear findings; organized process documentation	Organization map and Midterm structure set; this process documentation drafted in LaTeX

earlier on those days, or move those sessions to the evening. I chose waking earlier, with the evening as a fallback and three days as a last resort. Around the same time I reorganized my files into one folder per week, each with a short summary of its hypotheses, findings and open items, and I wrote that my failed experiments would be as much a part of the capstone as the results that work. By mid-October I wrote that the morning block had let me push forward at a consistent rate.

My Notion journal holds dated reflections from early September onward, and the Gantt chart holds the plan. Code, results and figures are backed up to a public GitHub repository, with every unit of work recorded in a work log. These records exist so that the failures stay visible, not only the results that worked.

Outcome Analysis

When I started in early September, I thought success would mean a better forecast, or better warnings before a low or a high. By late September, as described in Navigation, I doubted that this was possible, and the doubt held up: compared fairly, at the same rate of false alarms, every band I tried warned about as well as the others (Figure 3). So I stopped trying to beat other methods on a single score and started trying to understand the data well enough to say something true and new about it. For this stage, success means analyzing the data in depth, coming away with a clearer understanding of it, and producing findings that can be compiled into a scientific paper and submitted to a conference.

What Success Looks Like for This Stage

To plan around that goal, I looked at work like the work I want to produce. My first model was GLIMMER (Khamesian et al., 2026). Because I replicated it, I saw both the product and the process behind it: a clear question, a stated way of scoring forecasts, and a comparison with existing methods. My second model was work on conformal prediction (Romano et al., 2019; Vovk et al., 2005), which showed me how an uncertainty paper states what a band promises and then checks whether it keeps that promise. My third was the proceedings of the AAAI Conference on Artificial Intelligence, which I collected to see what

work at a venue I might submit to looks like, and which I am still reading. From these I took four outcomes, each written so that someone else could judge it. [TO CONFIRM: these four success criteria]

1. **I understand where forecast error comes from.** I can explain in plain terms when the forecast is wrong, for whom, and why, with evidence behind each answer. This matters most to me because deep understanding is what I wanted from the capstone in the first place.

2. **At least one result holds on data it was not designed on**, against a rule written before the test. This matters because a pattern found on the data I explored could be a pattern in my own choices rather than in the patients.

3. **The findings can carry a conference paper.** A reader can state the main claim in one sentence, see it compared with standard methods, and find its limits written down, as in the papers I modelled on. This is what turns the analysis into a deliverable.

4. **I catch my own errors and correct them in the open.** Each correction is logged as a new entry that names what it corrects, so anyone can see it. This is the part of the process I am proudest of, so I want it to be visible, not only claimed.

How the Work Measures Up

Understanding: mostly met. This is where I feel the biggest change. In September I could not have explained why a forecast fails. Now I can say that a standard band is honest on average but wrong in particular situations, that a patient is hard to forecast mostly because their glucose moves fast, and that much of the size of any single error looks like noise an hour ahead. I also understand why ideas borrowed from finance and weather did not give better warnings, the question that stopped me in late September.

What holds up: met. The situation-aware band was more even across situations than the standard band in every study I tested, including all eight outside studies, and a slow adjustment for each patient removed most of what was left. Both passed rules set before the tests, so I trust them more than anything I noticed while exploring.

Paper-ready findings: partly met. I have the claim, but not yet all the evidence a conference would want. The comparison with standard methods ran on only two datasets, most forecasters were trained with one seed per patient, so some results are preliminary, and nothing here measures whether an honest band helps a patient in practice.

Catching my own errors: met. The seed bug in my first weeks, described in Navigation, changed how I work more than any single result did. A result that looked like progress came from a mistake in my own code, and since then I check a result again before I build on it. That habit later caught wrong treatment labels for one group of patients and a baseline set up unfairly, and I logged each correction. I also learned that failed ideas are part of the work. About half of my tests failed, and keeping them on record is what makes me trust the ones that passed.

How Habits of Mind and Learning Outcomes Shaped the Work

I chose most of my HCs and LOs in a planning spreadsheet at the start of the semester, and I returned to it when preparing this deliverable. A few changed the work far more than I expected.

The biggest change came from #testability and the habits about my own biases, #biasidentification and #biasmitigation. #testability was not on my early sheet; I added it when I revisited the plan, because by then it had changed how I work. By mid-October I had many small results and no way to tell which were real, and after weeks on the bands I wanted them to work, which made it tempting to keep looking for a measure on which they looked good. Phrasing every idea so that it could fail, and fixing the rule for success before seeing the answer, became a written way of working, described in Use of Generative AI. It does not remove that wish, but it leaves the wish far less room to move the bar. The

second surprise was #studyreplication. I chose it because I was starting from GLIMMER, but repeating the study carefully is what exposed the seed bug, and it taught me that a replication tests my own code as much as the original paper.

My take-home message is that these habits mattered most when they became routines I could not skip on a busy day. Applied only when I remembered them, they would have caught far fewer of my mistakes. The list that follows shows where each one appears in the work.

Use of Generative AI

Claude Code, an AI coding assistant made by Anthropic, writes most of my code. I decide what to test and check what comes back. That split has held since my first week, when I wrote in my journal that I was working "more as a validator than anything else" while I replicated GLIMMER. What changed over the semester is how much the assistant does on its own. Until mid-October, it asked me before any run longer than 10 minutes. Then I wrote a research protocol and let the assistant work under it with no check-ins. The protocol is what makes that safe, so I describe it first.

The Research Protocol

The protocol, saved in the project repository, turns every idea into a test with the same steps.

1. **Every idea becomes a hypothesis.** It is written as one sentence about something that can be measured. "Does the patient explain more of the forecast error than the situation?" qualifies. "What drives uncertainty?" does not.

2. **The rule for success is written down and saved before any test runs.** The hypothesis goes into the register with the measure, the threshold that counts as success and the data it will use. The entry is saved in version control (git), which time-stamps every change, so anyone can check that the rule came before the result. The rule cannot move after the run. If I think of a better design after a failure, it

becomes a new hypothesis, labelled as a second or third try.

3. **The AI runs the smallest test that can answer it.** It reuses saved models and results wherever it can and trains only what the question needs.

4. **The result is judged only against the saved rule.** The verdict is supported, not supported or inconclusive. Anything noticed afterwards is labelled as noticed afterwards.

5. **Failed ideas stay in the register.** Nothing is deleted. A line of work stops after two failures in a row that suggest nothing new to test, and a method is never tuned until it wins. If it takes many tries, the number of tries is reported.

6. **A finding must hold on studies never used to build the method before it counts.** A result on one dataset is a lead. A result that holds on two or more datasets, and still holds when patients are resampled, is a finding. Only a result that also holds on the eight outside studies can be a main result of the paper.

7. **Everything is logged.** Each unit of work gets an entry in a work log with links to the code and result files behind it, and each week has a short summary of what was found and what is still open.

The protocol also carries the statistical rules I had learned the hard way: the patient, not the reading, is the unit of analysis; test data stays out of fitting and calibration; every model and patient gets its own random seed; and methods are compared at a matched width or a matched false-alarm rate, never by which one is wider. Since then the assistant has run dozens of tests this way. Writing the rules down is what made it safe to let it work without asking me each time.

What the AI Did and What I Decided

The assistant wrote the data loaders, the forecasting models, the uncertainty methods and the scripts that score them. It ran the training jobs, saved the results to files

and wrote a first reading of each result. I chose the topic, the datasets and the questions, I set what counts as success, and I made the calls to change direction, such as closing the autoencoder line in early October. Table 2 summarizes the split.

Table 2

Who Did What in the Project

Task	Claude Code	Me
Choosing questions	Proposes next questions under the protocol (since mid-October)	Chose the topic, the datasets and every change of direction
Defining success	Writes the rule into the register before a run	Wrote the protocol that requires a rule before every run
Writing code	Wrote most of the code	Read through all of it and asked for checks
Running analyses	Ran training jobs and saved results	Approved long runs until mid-October
Judging results	Gives a first verdict against the rule	Read the figures and made the final call
Keeping records	Writes work log entries	Daily summaries and the Notion journal

How I Checked the AI's Work

An assistant that writes code quickly can also produce wrong results quickly, and a wrong result can look convincing. My first check is simple: I read through all the code the assistant writes. I do not accept a result from code I have not read. Beyond that, I relied on four safeguards on top of the protocol. First, I wrote the rules the assistant works under into a file called `CLAUDE.md`, which it reads at the start of every session. Second, every number in my writing must come from an output shown in a session or from a saved result

file. Third, the work log records every unit of work, and a mistake is fixed with a new correction entry that names the entry it corrects, rather than by rewriting the old one. Fourth, `CLAUDE.md` holds a list of traps already hit: a shared random seed, test data leaking into fitting, and one patient making up 53% of the readings in a pooled dataset. It also requires that results from one seed or five patients be called preliminary. These rules apply to baselines too. In mid-October a first run showed that GARCH, used to predict how large each error would be, was off by a constant, which made it look worse than it is, so it was rescaled on validation data before any comparison, rather than beaten in its weak form. [TO FILL: how I read the code in practice, for example what I look for first and one thing reading it caught]

Feedback and How I Used It

Prof. Watson's written feedback on my early proposal made three research suggestions. I treated each one as a question to test, with a rule for success written down first. Below, each suggestion is followed by what I did and what it showed.

Approximate Entropy to Model Major Shifts

The suggestion. Prof. Watson suggested exploring approximate entropy (Pincus, 1991), a measure of how irregular a series is, to model the major shifts in a patient's glucose.

What I did. In early October I tested sample entropy (Richman & Moorman, 2000), a close relative of approximate entropy that is less sensitive to the length of the record. I asked whether a patient's sample entropy, measured on the first part of their data, could predict how large their forecast errors would be later, beyond what the plain spread of their glucose already predicts. Before running, I set the rule that the measure had to cut the prediction error by at least 15%.

What it showed. Sample entropy added only a little: 9% beyond how variable a patient's glucose is, short of the 15% I had set. This was on the 25 AZT1D patients with one training seed, so it is preliminary. I recorded it as a hint, not a finding, and did not

build on it.

Autoencoders

The suggestion. Prof. Watson also suggested autoencoders, networks that compress an input into a few numbers and then rebuild it, so that the few numbers act as a summary.

What I did. I trained an autoencoder that compresses two hours of readings into a short summary, first on AZT1D and then on HUPA-UCM (Alvarado, 2024). I first drew the summaries with PCA, a method that keeps the directions along which the data varies most. Then, at Prof. Watson's suggestion, I drew them with t-SNE instead, a method that places similar points close together on a flat map and so shows groups that PCA can hide. I also tried sizing the bands by groups learned from the summaries instead of by my hand-drawn situation grid.

What it showed. The autoencoders mostly captured glucose level and direction. With PCA, the main direction of the summary tracked current glucose almost exactly. The t-SNE maps sorted the windows by current level and then by trend, the same structure as the situation grid I was already using (Figure 8). Bands built from the learned groups did no better than the hand-drawn grid. Because the autoencoder showed nothing beyond level and trend, I closed this line in early October. When I returned to it in mid-October with a stricter test, level and trend explained 87% of what the autoencoder kept, which confirmed that decision.

Separating Meals and Insulin From Changes That Are Not Externally Predictable

The suggestion. Prof. Watson noted that most major glucose changes come from outside factors such as eating and insulin, which should be filtered out, and that subtler features producing major changes, ones that are not externally predictable, would be really good to know about.

What I did. I took the part of the autoencoder summary that level and trend did not explain and asked which recorded variables explained it for patients the model had not

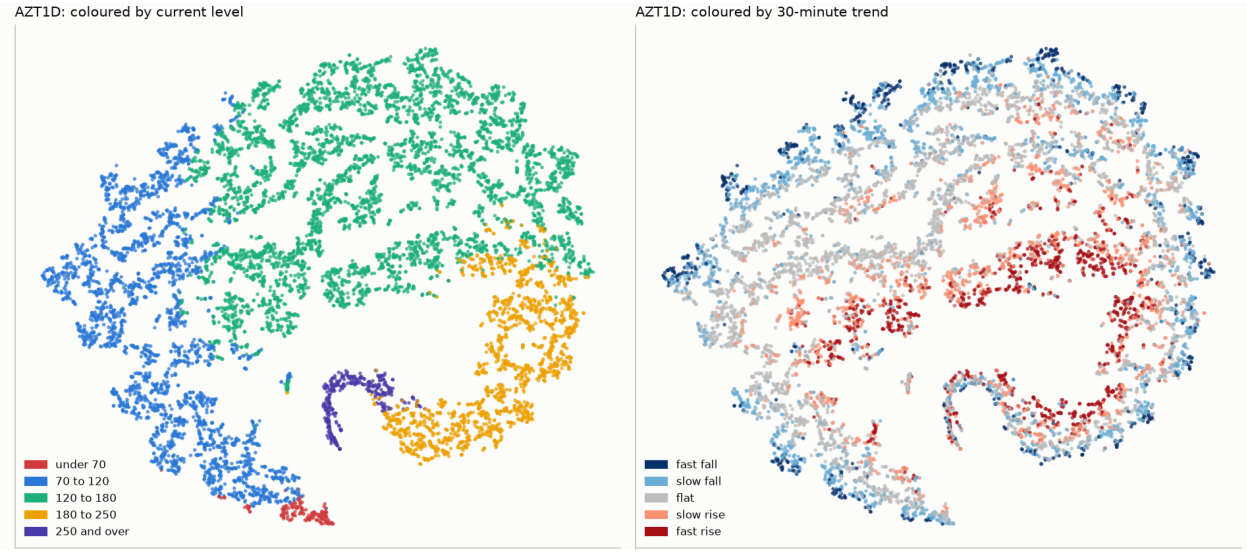


Figure 8

The autoencoder learned glucose level and direction. Each dot is one two-hour AZT1D window on a t-SNE map, coloured by its current glucose level (left) and by its 30-minute trend (right).

seen. The candidates included carbohydrates, insulin doses, insulin still active in the body, basal insulin, heart rate, steps and time of day. Later, across all 215 patients from 11 studies, I asked what makes some patients harder to forecast than others.

What it showed. Meals and insulin explained almost none of what was left. The leftover tracked mainly the patient and their basal insulin setting, and it did not help predict how wrong a forecast would be. The subtler driver turned out to be how fast a patient's glucose moves. The spread of each patient's 30-minute glucose changes explained 59% of the differences in forecast difficulty in studies left out of the fit, against 44% for the overall spread of their glucose. It also accounted for children and closed-loop users being harder to forecast: once it was included, neither group differed clearly from the others. This answer came from following his third suggestion, and it is now one of the findings in the paper.

Habits of Mind and Learning Outcomes

The entries below are not a checklist of everything I did. Each one shows where a learning outcome changed how I worked or what I produced, and points back to the section where that moment is described.

Capstone Learning Outcomes

#navigation Navigation meant planning around how I actually work and changing course when a plan stopped serving the project. My planning went through three stages, described in Planning: checklists in Notion, then a hand-drawn week-by-week plan on my iPad, then a Gantt chart built around fixed commitments: my second-reader meeting, this Midterm and the Second Presentation. To protect the work from distraction I set a morning block for the capstone. When I noticed I was skipping it, I chose a backup in advance: wake earlier, move to the evening if that failed, and accept three days a week only as a last resort (Working Habits and Tools). The hard part of setbacks was knowing when to switch. I set recursive forecasting aside once it was clear it could not predict meals, moved from forecasting to uncertainty when accuracy looked crowded, closed the autoencoder line when it showed nothing beyond level and trend, and dropped the simulated patients when they proved too easy to forecast (Navigation). Later, the protocol's stopping rule (Use of Generative AI) made that call less a matter of mood. I treated each of Prof. Watson's suggestions as a question to test (Feedback and How I Used It), and I came to my second-reader meeting with slides ready. Table 1 is the honest record: the research often ran ahead, but the documentation fell behind. I planned to finish it in mid-October and drafted it in late October, a reminder that the planning fallacy applies to writing as much as to research.

#outcomeanalysis I tried to decide what success looks like before doing the work, at two levels. For this stage, the Outcome Analysis section states what success means now

and why it changed. Once better warnings looked out of reach, success became understanding the data in depth and producing findings strong enough to anchor a paper. These outcomes matter most to me because they match what I am proudest of: catching my own errors and understanding the material. I wrote the criteria so that someone else could check them, for example whether a result still holds on studies that were never used to build it. Existing work shaped them. Replicating GLIMMER showed me what a published study in this area does and where it can break, and I collected AAAI proceedings to see what current research there looks like and which directions it values (Next Steps). For single tests, every question had its rule for success written before the run, as described in Use of Generative AI, and I judged the verdict only against that rule. This changed the work. When sample entropy came up short of the bar I had set, I recorded it as a hint rather than looking for a measure on which it would pass. I also chose a set of HCs and LOs to apply at the start of the semester. [TO FILL: looking back at that plan now, which HCs and LOs actually shaped the work, which I dropped or added, and why]

#qualitydeliverables The Handbook asks what I want to proudly show off after a year of work. My answer is the two things I named in the Overview: catching my own errors and the understanding of the material I built. So I judged quality by whether I can stand behind a result, not by how good it looks. That meant comparing my method against strong standard methods rather than weak ones, repairing a baseline before comparing against it, and saying plainly in Outcome Analysis where my method does not win. My real-world models are published studies, starting with GLIMMER, and the conference papers I am now reading to judge what a competitive paper needs. The Handbook notes that skillful use of AI can let a capstone reach a larger scope than one done in 2020, as long as that use is documented. That is the case here. With an AI writing most of the code, I could test many more ideas on many more patients than I could have managed by hand, and the Use of Generative

AI section documents how I kept that larger scope honest. I think the depth fits the midterm stage. The research is broad and checked on data it was not designed on, and the paper draft has its methods and approach written. What is not yet at the level I want is the paper itself: turning the findings into one clear argument is the work that remains (Next Steps).

#curation Most of what I did is not in this document, and that was a choice. The hypothesis register and the work log hold every test, including many side analyses, and I left them there because the Midterm asks how I produced this stage of the work, not for a catalog of everything I tried. From it I selected the turns that changed my direction and the corrections I learned the most from. I also left out the statistical detail of most tests, because it belongs in the paper and its supplement. I chose each figure to explain one idea: what a warning is (Figure 1), how forecast errors are judged (Figure 2), why a wider band mostly adds alarms (Figure 4), and what the chosen direction fixes (Figure 5). I ordered the document so that terms such as hypoglycemia and the Clarke grid are defined before they are used, and so that the research protocol is explained once, in Use of Generative AI, and only referred to elsewhere. The document also changed through review. An earlier draft had separate sections on setbacks and on modeling a professional process, and annotated examples of AI use. I cut them after feedback, and the setbacks now appear in Navigation at the point in the story where each one happened. Behind the document, the work is organized so it can be found: one folder per week with a short summary, the work log, and a catalog of every dataset with its source and license.

Major Learning Outcomes

#cs156-MLCode The project rests on several machine learning systems written in Python: per-patient CNN-LSTM forecasters that read recent readings and predict glucose an hour ahead, autoencoders, deep ensembles, and a quantile network that

predicts its own band. The AI wrote most of this code, and I read all of it, as described in Use of Generative AI. Reading it taught me that working is a weak standard. The seed bug in the GLIMMER replication ran without a single error and produced a believable gain, yet every per-patient search was identical (Navigation). Working code can still be wrong, so every model and every patient now gets its own seed by rule. Performance mattered too. Training one forecaster per patient across many studies stayed manageable because saved models and results are reused rather than retrained for each new question, and long training jobs can resume where they stopped.

#cs156-MLExplaination I had to explain what the models learned, not only how well they scored. For the autoencoder, I drew its summaries with PCA and then with t-SNE, and the t-SNE map showed in one picture that it had sorted the windows by glucose level and then by trend (Figure 8). That picture is what let me close the line with confidence. In writing, this document introduces each band method in one plain clause, such as GARCH as a volatility tracker from finance and conformal prediction as a way to size the band from past errors, and it describes an honest band as one that contains the true value as often as it claims. Orally, I explained the project in slides for my second-reader meeting in early October, including my doubt about whether the direction was valuable.

#cs166-EmpiricalAnalysis My main simulation used patients from simglucose, an open-source type 1 diabetes simulator, with sensor noise of a known size planted on top. Because the planted noise was known, it could act as an answer key: a good band should widen by about as much noise as I added. The results taught me two things. The answer key itself was too simple: noisy inputs also made the forecasts worse, so the error grew by more than the noise I planted. And the simulated patients were far easier to forecast than real ones, so the simulation could not answer

my question (Navigation). I also used simulation in a second way, re-creating the data as if my idea were false. I shuffled the situation labels, and later shifted them in time within each patient, many times, to see how good or uneven results look by chance alone. A result counted only if it stood clear of that chance picture.

#cs114-ModelSelection Several of my questions came down to which explanation best fits the data. When forecast errors varied, I asked whether the situation, the patient or the day accounted for more of it, and compared these accounts directly rather than assuming one. The situation explained more of how large an error would be, while most of the unevenness left in the bands was between patients, which is why the final band has one adjustment for each (Outcome Analysis). When comparing band methods, I compared them at a matched false-alarm rate, because a wider band always looks safer (Figure 3). Before calling any unevenness real, I compared it with a chance-only model, the unevenness that sampling alone produces. This changed my reading more than once: a method that looked more even was mostly less noisy, not better calibrated, and some unevenness that looked like a failure was no larger than chance.

#cs110-CodeReadability I organized the code so that someone else could read and run it. Shared code lives in one library that stays in place, while each week's scripts and notebooks sit in that week's folder, which separates reused code from code for one experiment. Project rules fix how every script and notebook finds the repository root, so no path breaks when a file is copied into a new week. When I moved the notebooks into weekly folders, I checked that each one still found its files from its new place. Conventions for names, comments and file placement are written in the instructions the AI reads at the start of each session, and the data loaders stop with a message that says what is missing rather than failing silently. Everything except the raw data, which is not ours to share, is in a public GitHub repository, and the

dataset catalog tells a reader where to get each dataset.

Habits of Mind

#hypothesisdevelopment Most of my hypotheses grew from an observation. In late September the bands were honest on average but not in every situation: the same 80% band covered too little in some situations and too much in others (Figure 5). The mechanism I proposed was that how hard glucose is to predict depends on where it is and where it is heading, so one band width is too narrow in some situations and too wide in others. GARCH, borrowed from finance, rests on a related idea: the size of errors changes with conditions. I also wrote down the main alternative, that the patient rather than the situation explains how large the errors are; the situation explained more on most datasets. Later, when some patients proved harder to forecast, I looked for a mechanism rather than a label. If a patient's glucose moves faster, more can change within an hour, so their errors are larger. That one mechanism also explained why children and closed-loop users seemed harder to forecast, against the alternative that the groups themselves were the cause.

#testability I tried to make each hypothesis produce several predictions, chosen so that a competing explanation would predict something different. For the situation-aware band, the first prediction was that it would be more even across situations than GARCH. That alone could not rule out a duller explanation: splitting the data into many groups helps, whatever the groups are. So a second test shuffled the situation labels and predicted that the real situations would beat every shuffled version. A third predicted that the gain would hold for a second forecasting model, and a fourth that it would hold on studies never used to design the method. Each had its own rule, set before the run, as described in Use of Generative AI. I also asked whether a test was practical, not only possible in principle. The simulated patients failed that check, because they were too easy to forecast to be informative. Prof. Watson's idea

about subtler drivers of glucose change was hard to test as stated, so I turned it into a narrower, measurable question.

#gapanalysis I first asked what already existed. Accuracy-focused forecasting is crowded, so a slightly better forecaster was unlikely to be my contribution. For bands, I started from existing methods rather than from scratch: GARCH from finance, the analog ensemble from weather and conformal prediction from statistics. Each is well tested in its own field, but at a matched false-alarm rate they performed almost identically, and none made better alarms than a tuned threshold on the forecast itself. When I built the situation-aware band, a literature check showed that it already had a name, Mondrian conformal prediction (Vovk et al., 2005), and I compared it with strong alternatives, deep ensembles (Lakshminarayanan et al., 2017) and conformalized quantile regression (Romano et al., 2019). The methods needed adapting, not reinventing. The gap I found is in how bands are judged and used: in my searches I found no glucose forecasting study that checks whether bands stay honest by situation, by patient and by day. The closest, an intensive-care study, reported coverage by glucose range. I record this as not found, not as proof that none exists.

#studyreplication I began by replicating GLIMMER (Khamesian et al., 2026), a recent study from the group that built AZT1D, before trying to extend it. This was a partial replication. I kept its core design: one CNN-LSTM per patient, a loss that weights errors in dangerous ranges more heavily, and a per-patient search for those weights. I changed two things to test how far its results generalize: I added a second model type, a Transformer, and a second dataset, OhioT1DM (Marling & Bunescu, 2020), with different patients. The main effect replicated on both datasets: the weighted models caught more dangerous events at the cost of overall accuracy. The per-patient gain did not survive once each search had its own seed, as described in Navigation. A failed replication can mean the original claim is wrong, my methods

are weaker, or my version was not faithful enough, and I cannot yet tell which. It does mean the personalization claim should not be relied on without further replication. That lesson shaped the research program: a result counts only once it holds on studies never used to build the method.

#dataviz I tried to match each plot to the question it answers. To show what a forecast error means clinically, I used the Clarke error grid, a scatter of forecast against actual glucose with zones drawn by clinical harm (Figure 2). To ask what widening a band does, I plotted the share of forecasts in each zone as the band widens (Figure 4), which showed that wider bands mostly add false alarms. Coverage across situations is one value for each pair of level and trend, so I used a heat map with a color scale centered on 80%, where cells that are too narrow and too wide stand out in opposite colors (Figure 5). For the autoencoder I switched from PCA to t-SNE at Prof. Watson's suggestion and colored the same map by level and by trend (Figure 8). Because t-SNE distorts distances, I read it only as a check on grouping, not as a measurement. I wrote captions that state the finding, such as "every method lands on nearly the same curve" (Figure 3), and I read each figure before writing up a result.

#confidenceintervals Two kinds of interval run through this work. The first are the bands. An 80% band is a prediction interval for one future reading rather than a confidence interval for a fixed value, but its claim works the same way: it is about the procedure. Across many forecasts, 80% of such bands should contain the true value, so I check coverage on held-out data by situation and by patient rather than trusting the stated level. The second are confidence intervals for my results. Readings from one patient are not independent, so I build these by resampling whole patients or whole studies, because resampling single readings would make them too narrow. For example, children were harder to forecast than adults after accounting for glucose variability, with a 90% interval of 3% to 12% higher error. The correct reading is that

intervals built this way would contain the true difference in 90% of repeated samples, not that there is a 90% chance this one does. It excludes zero, yet the difference faded once I accounted for how fast glucose moves.

#sampling The population I care about is people with type 1 diabetes who wear a glucose sensor, and my unit of analysis is the patient, not the reading. One patient contributes thousands of closely related readings, so treating each reading as independent would be pseudoreplication and would overstate the evidence. This matters in practice: in HUPA-UCM one patient makes up over half of the readings, so a pooled result there could describe one person. I therefore report results patient by patient and resample patients when estimating uncertainty. For representativeness, I chose outside studies that differ in age, insulin delivery and sensor: children and adults, closed-loop and other pumps, and three sensor brands. Within each, patients were drawn in random order from those with enough weeks of data, and those studies were held out from design. The sample still has limits. Everyone in it joined a research study, so it may not reflect all people with type 1 diabetes, and results from one seed or a few patients are labelled preliminary.

#biasidentification The biases I had to watch most were my own. After weeks invested in uncertainty bands, I wanted them to work. That is the sunk cost effect: past effort makes it harder to accept that a path is not paying off. It feeds confirmation bias, the tendency to seek and accept evidence for what I already hope. When the bands failed my first test in late September, the two together made it tempting to hunt for a measure on which they looked good. The seed bug showed a second interaction. The tuned models seemed to lower the error, which matched what the published study led me to expect. The code ran without errors, and I leaned on a shortcut: working code is correct code. Because the result agreed with my expectation, nothing prompted me to question the shortcut. The planning fallacy shows in my timeline. I planned to

finish this documentation by mid-October, and it slipped, which fits the pattern of judging a task by its best case rather than by how long similar work took before.

#biasmitigation I cannot remove these biases, only make them harder to act on. Against confirmation bias, I write the rule for success before each test runs, as described in Use of Generative AI. A fixed bar leaves no room to pick a friendlier measure afterwards. Comparing methods at a matched false-alarm rate works the same way: a wider band cannot look better just by warning more. Against the working-code shortcut, I read all the code the assistant writes and re-check a result before relying on it, most of all when it looks good, since good news is what I least want to question. Against the sunk cost effect, I decided early in October that failed experiments are as much part of the capstone as successes, and I keep every one on record. A failure then counts as a result rather than wasted effort, which weakens the pull to keep going on a path that is not paying off. Against the planning fallacy, my plan-versus-actual table gives me a record of how long work really took. Bias can still enter through which questions I choose to ask.

#organization I ordered the content so that each idea comes after what it depends on. The Overview gives the roadmap: the question, why it matters to a person with diabetes, and what the later sections cover. Terms such as hypoglycemia, the band and the Clarke grid are defined when they first appear, before any result uses them. Navigation is ordered by the turns in the project, with subsections named mostly for changes of direction, so a reader follows why the work moved and not only when. In Feedback, each suggestion has the same three parts, the suggestion, what I did and what it showed, so the entries can be compared. Every figure and table is referenced in the text where it supports a point. I considered two other structures. A strict chronological diary would match my journal but bury the decisions in daily detail. A section per hypothesis would be complete but lose the story that connects them. The

paper is organized differently, around its findings, with each results heading stating what was found.

#professionalism I followed the established conventions of academic work. This document is written in LaTeX in APA 7 style, with in-text citations and a reference list, within the structure the Midterm instructions set. I checked each reference against its source, which showed that I had credited MetaboNet to the wrong authors, and I corrected it. For data, I followed each source's access process. OhioT1DM required an access request, and I followed the form its maintainer sent before using it. I record every dataset's source, licence and terms of use in a catalog, and because some of the data are licensed for research use and are not ours to redistribute, the raw files stay out of my public repository. I credit borrowed methods by their standard names, such as Mondrian conformal prediction, rather than presenting them as my own. I state how I used generative AI in its own section, with a table of who did what. [TO FILL: how I proofread this document and who else read it]

#audience My first readers are Prof. Watson and the capstone committee. They know machine learning and statistics but not diabetes, so the risk is not the methods but their clinical meaning. Writing for clinicians instead would have assumed the clinical terms and compressed the methods, the reverse of what these readers need. I therefore define hypoglycemia, hyperglycemia and the Clarke grid before using them, and explain each borrowed method in one plain clause. I describe a band's quality in terms a reader can check and keep numbers to the few that carry a point. I also anticipated a likely misreading, that a wider band is better because it catches more, which is why I compare methods at a matched false-alarm rate. I gathered input from this audience as I went: Prof. Watson's written feedback shaped three lines of work, and his suggestion to use t-SNE made the autoencoder figure clearer. [TO FILL: how I tailored the slides for my second-reader meeting to that audience] My later audience

is conference reviewers, so I collected AAAI proceedings to learn what they expect.

Next Steps Toward the Full Draft

I now have a wide body of research: 85 hypotheses in the register across 11 studies, with a main result that holds on eight studies never used to design it. What I do not yet have is one clear paper. The next stage is to shape this work into a single argument and to decide which results belong in it and which stay in the supplement.

The capstone requirements may already be met by this work, but a publishable paper may need expanded research. The current version of the paper has the research methods, an explanation of the approach, and the use of the forecasting model to create bands that stay consistent with their stated coverage across situations and patients. What remains is to write the results and discussion around the main finding: an honest band needs a fixed adjustment for the situation and a slow adjustment for each patient, learned while the band is in use.

I may also expand the research in a different direction. I collected proceedings from the AAAI Conference on Artificial Intelligence to see what current research there looks like and which directions it values. After reading them, I will decide whether the present framing fits, or whether the paper needs an extension to be competitive there. [TO FILL: which direction I take after reading the AAAI proceedings, whether I target AAAI or another venue, and what I think its reviewers will ask for first]

Several practical items remain before the Full Draft, which is due December 11, 2026:

- Audit every number in the paper against its result file, and bring the paper supplement up to date with the hypothesis register, which has grown since the supplement was made.
- Restore the MetaboNet file and, if time allows, run the two optional checks it blocks: comparing my bands with deep ensembles on OhioT1DM, and repeating the analysis

on a larger panel of up to 60 patients per held-out study. The paper does not depend on either.

- Move the paper itself into LaTeX, as I did for this document, since LaTeX handles the structure of a paper better than Google Docs and is easier to work on with the AI.

The Second Presentation, the go or no-go review, is on November 20. By then I want the direction of the paper settled, so that the remaining weeks go to writing rather than new research. [TO FILL: what I want to have ready for the November 20 presentation]

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